

CCNA (200-301)



COURSE CURRICULUM

Network Fundamentals

- Explain the role and function of network components
- Routers
- L2 and L3 switches
- Next-generation firewalls and IPS
- Access points
- Controllers (Cisco DNA Center and WLC)
- Endpoints
- Servers

Describe characteristics of network topology architectures

- 2 tier
- 3 tier
- Spine-leaf
- WAN
- Small office/home office (SOHO)
- On-premises and cloud

Compare physical interface and cabling types

- Single-mode fiber, multimode fiber, copper
- Connections (Ethernet shared media and point-to-point)
- Concepts of PoE

Identify interface and cable issues (collisions, errors, mismatch duplex, and/or speed)

Compare TCP to UDP

Configure and verify IPv4 addressing and subnetting

Describe the need for private IPv4 addressing

Configure and verify IPv6 addressing and prefix

Compare IPv6 address types 1.9.a Global unicast

- Unique local
- Link local
- Anycast
- Multicast
- Modified EUI 64

Verify IP parameters for Client OS (Windows, Mac OS, Linux)

Describe wireless principles

- Nonoverlapping Wi-Fi channels
- SSID
- RF
- Encryption

Explain virtualization fundamentals (virtual machines)

Describe switching concepts

- MAC learning and aging
- Frame switching
- Frame flooding
- MAC address table

2.0 Network Access

- Configure and verify VLANs (normal range) spanning multiple switches
- Access ports (data and voice)
- Default VLAN
- Connectivity

Configure and verify interswitch connectivity

- Trunk ports
- 802.1Q
- Native VLAN

Configure and verify Layer 2 discovery protocols (Cisco Discovery Protocol and LLDP)

Configure and verify (Layer 2/Layer 3) EtherChannel (LACP)

Describe the need for and basic operations of Rapid PVST+ Spanning Tree Protocol and identify basic operations

- Root port, root bridge (primary/secondary), and other port names
- Port states (forwarding/blocking)
- PortFast benefits

Compare Cisco Wireless Architectures and AP modes

Describe physical infrastructure connections of WLAN components (AP, WLC, access/trunk ports, and LAG)

Describe AP and WLC management access connections (Telnet, SSH, HTTP, HTTPS, console, and TACACS+/RADIUS)

Configure the components of a wireless LAN access for client connectivity using GUI only such as WLAN creation, security settings, QoS profiles, and advanced WLAN settings

3.0 IP Connectivity

Interpret the components of routing table

- Routing protocol code
- Prefix

- Network mask
- Next hop
- Administrative distance
- Metric
- Gateway of last resort

Determine how a router makes a forwarding decision by default

- Longest match
- Administrative distance
- Routing protocol metric

Configure and verify IPv4 and IPv6 static routing

- Default route
- Network route
- Host route
- Floating static

Configure and verify single area OSPFv2 3.4.a Neighbor adjacencies

- Point-to-point
- Broadcast (DR/BDR selection)
- Router ID

Describe the purpose of first hop redundancy protocol

4.0 IP Services

- Configure and verify inside source NAT using static and pools
- Configure and verify NTP operating in a client and server mode
- Explain the role of DHCP and DNS within the network
- Explain the function of SNMP in network operations
- Describe the use of syslog features including facilities and levels
- Configure and verify DHCP client and relay
- Explain the forwarding per-hop behavior (PHB) for QoS such as classification, marking, queuing, congestion, policing, shaping
- Configure network devices for remote access using SSH
- Describe the capabilities and function of TFTP/FTP in the network

5.0 Security Fundamentals

- Define key security concepts (threats, vulnerabilities, exploits, and mitigation techniques)
- Describe security program elements (user awareness, training, and physical access control)
- Configure device access control using local passwords
- Describe security password policies elements, such as management, complexity, and password alternatives (multifactor authentication, certificates, and biometrics)
- Describe remote access and site-to-site VPNs
- Configure and verify access control lists
- Configure Layer 2 security features (DHCP snooping, dynamic ARP inspection, and port security)
- Differentiate authentication, authorization, and accounting concepts
- Describe wireless security protocols (WPA, WPA2, and WPA3)
- Configure WLAN using WPA2 PSK using the GUI

6.0 Automation and Programmability

- Explain how automation impacts network management
- Compare traditional networks with controller-based networking

- Describe controller-based and software defined architectures (overlay, underlay, and fabric)
- Separation of control plane and data plane
- North-bound and south-bound APIs

Compare traditional campus device management with Cisco DNA Center enabled device management

Describe characteristics of REST-based APIs (CRUD, HTTP verbs, and data encoding)

Recognize the capabilities of configuration management mechanisms Puppet, Chef, and Ansible

Interpret JSON encoded data



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